Jiaqi Tu

SUMMARY

Data Engineer with 3+ years of experience designing, implementing, optimizing, and supporting enterprise data platforms. Expert in AWS cloud services, Databricks, and modern data architectures, with hands-on experience building configuration-driven ETL frameworks and multi-source ingestion pipelines. Adept at translating business requirements into governed, high-quality, analysis-ready data products. Experienced in Customer 360, MDM, slowly changing dimensions, metadata management, data quality, and performance tuning using Python, PySpark, and SQL. Familiar with leveraging multiple AI providers to build a personal AI agent network that breaks through individual productivity limits.

TECHNICAL SKILLS

- Cloud & Big Data: Amazon S3, Amazon Redshift, AWS Glue, AWS Step Functions, AWS Lambda, AWS Transfer Family, DynamoDB, EventBridge, CloudFormation, AppFlow, Amazon SageMaker, CodeCommit, AWS Secrets Manager, IAM, AWS Systems Manager, Databricks
- Programming: Python, PySpark, SQL, Bash/Shell scripting, C++, C
- Data Engineering: ETL/ELT, data ingestion, data modeling, dimensional modeling, SCD, MDM, Customer 360, data quality, metadata management
- Databases & Warehouses: Amazon Redshift, Microsoft SQL Server, PostgreSQL, DynamoDB
- AI & Agents: multi-provider LLMs, personal AI agent network, agent orchestration, GitHub Copilot, OpenAI Codex
- Tools & Delivery: Git, Jenkins, CI/CD, REST APIs

WORK EXPERIENCE

Data Engineer | 407 ETR | May 2023 - Present

- Design, build, and maintain reusable, configuration-driven ETL pipelines that ingest structured, semi-structured, and unstructured data from diverse source systems into AWS-based data platforms.
- Translate business, analytics, data governance, and data quality requirements into scalable ingestion patterns, validation rules, data cleansing logic, and curated datasets for downstream analysis.
- Develop cloud-native workflows using Amazon S3, AWS Glue, AWS Step Functions, AWS Lambda, EventBridge, AppFlow, AWS Transfer Family, DynamoDB, and Amazon Redshift, selecting services based on source, latency, security, and operational requirements.
- Implement and optimize slowly changing dimension strategies, including Type 1 and Type 2 patterns, and support Customer 360 and MDM solutions by integrating cross-domain customer data into trusted, reusable data products.
- Optimize PySpark, Python, SQL, AWS Glue, and Amazon Redshift workloads through efficient transformations, partitioning, incremental processing, query tuning, and appropriate resource sizing.
- Support production pipelines through monitoring, troubleshooting, and root-cause analysis; apply IAM, AWS Secrets Manager, AWS Systems Manager, CloudFormation, Git, Jenkins, and CI/CD practices to deliver secure, repeatable solutions.
- Support the AWS-to-Databricks platform migration by assessing existing pipelines, contributing to migration planning and validation, and implementing Bronze, Silver, and Gold data layers aligned with medallion architecture.
- Designed and deployed end-to-end, configuration-driven machine learning pipelines using Amazon SageMaker and AWS Glue, including data preprocessing, feature store management, model training, and validation to support scalable ML workflows.

Academic Tutor | Easy Group | July 2022 — December 2022

- Tutored students in mathematics and electrical engineering, reinforcing classroom concepts and resolving complex academic problems. Guided students through comprehensive course structures to improve their understanding of core materials.

EDUCATION

Bachelor of Applied Science | University of Toronto | Toronto, ON | 2021 - 2025
Specialization: Software Engineering | Minor: Artificial Intelligence

CONTACT

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